

CSE 2215: Data Structure and Algorithms-I

1. Show the mechanism of the following sorting techniques to sort the given data in descending order.

- i) Counting Sort
- ii) Insertion Sort
- iii) Quick Sort
- iv) Merge Sort

Given Data: 6 4 3 5 2

2. How many times the condition of while loop in insertion sort (ascending) will be executed for the following data? *With Mechanism*

Set-I: 10, 20, 30, 40 (best case)

Set-II: 40, 30, 20, 10 (worst case)

Set-III: 30, 10, 40, 20 (Average case)

3. Discuss the best case, worst case, average case time complexity of insertion sort algorithm.

4. How many times each of statements and conditions will be executed?

Handwritten annotations:
- Above `k=0;`: $1 \rightarrow 4$
- Above `for (i=0; i<=3; i++){`: $3 - 0 + 1 + 1$
- Above `sum=0;`: 4
- Above `for (j=1; j<4; j++){`: 4
- To the right of the code block: (n)
- Below the code block: $\square$

```
k=0;
for( i=0; i<=3; i++){
    sum=0;
    for (j=1; j<4; j++){
        sum=sum+i+j;
        k=k+1;
    }
    k=1;
}
```

5. Explain the time complexity of counting sort, average case time complexity of Quick sort, and time complexity of merge sort.

6. Draw a recursion tree for the ascending order merge sort using the following data.

78 23 56 26 89 37 28

Handwritten: slide

7. How many element comparisons are needed for the following instance of ascending order quick sort algorithm to find the first partitioning element and second partitioning element?

18 23 56 26 89 37 28 48

8. Write algorithms for descending order i) counting sort ii) insertion sort iii) quick sort iv) merge sort.